London Business School

Mergers, MBOs & Other Corporate Reorganizations

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Final Review

Q1(i)

- True or False: The case of Roche shows that in times of crisis it is impossible to raise debt; therefore firms should raise equity.

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- True or False: The case of Roche shows that in times of crisis it is impossible to raise debt; therefore firms should raise equity.

- Solution: False. The Roche–Genentech case shows that although bank lending was restricted, the bond market still provided financing. Capital markets tend to be less procyclical than bank lending.

Q1(ii)

- True or False: In Chapter 11 the absolute priority rule can never be broken.

- Solution: False. In the LATAM case, new senior debt was proposed that would rank ahead of existing debt to solve a debt overhang problem.

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Q1(iii)

- True or False: Excess returns from asset manager engagements indicate insider information and violate securities laws.

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- Solution: False. The information is soft information obtained through engagement and fits under the mosaic rule, which is permissible.

Q1(iv)

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- Solution: False. Regulation can limit private benefits, and blockholders can reduce agency costs associated with dispersed ownership.

Q1(v)

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- True or False: Lenders should only give haircuts to borrowers that are insolvent.

- Solution: False. Haircuts may also be given to distressed but solvent firms to resolve debt overhang and enable profitable investments.

Q2(a)

- Acquirer Plc has plans to acquire Target Corp; both companies are all-equity financed. Preacquisition (before any merger information or rumors are circulating), Acquirer is trading for $40 per share and has 200 million shares outstanding. Target is trading for $30 per share, with a pre-merger value of $4.5 billion. Assume that the present value of projected synergies is $1 billion and that the deal will be financed with cash (i.e., cash merger). The board of Target indicated that it would sell the firm if it is offered $5 billion in cash.

- i) What is the value of Acquirer after the acquisition? Should Acquirer acquire Target? What is the NPV of the deal for Acquirer?

Q2(a)

- Acquirer value before merger = 200m × $40 = $8bn. Target value = $4.5bn. Synergies = $1bn. Offer = $5bn.

- Solution:

Q2(a)

- Acquirer value before merger = 200m × \$40 = \$8bn. Target value = \$4.5bn. Synergies = \$1bn. Offer = \$5bn.

- Solution:
- Post-merger value of Acquirer = 8 + 4.5 + 1 − 5 = \$8.5bn.
- NPV for Acquirer = 8.5 − 8 = \$0.5bn.
- Decision: Acquire Target.

Q2(b)

- Find Acquirer share price after acquisition and premium paid.

- Solution:

Q2(b)

- Find Acquirer share price after acquisition and premium paid.

- Solution:
- Post-merger share price = 8.5bn / 0.2bn shares = $42.5.
- Premium = (5 − 4.5)/4.5 = 11.11%.

Q2(c)

- Maximum price Acquirer could pay.

- Solution:

Q2(c)

- Maximum price Acquirer could pay.

- Solution:
- Maximum payment = Target value + Synergies = 4.5 + 1 = $5.5bn.
- Any price below $5.5bn keeps the acquisition profitable.

Q2(d)

- Stock swap merger: maximum exchange ratio.
- Solution:

Q2(d)

- Stock swap merger: maximum exchange ratio.

- Solution:
- Target value with synergies = $5.5bn.
- Acquirer share price = $40.
- Shares issued = 5500/40 = 137.5m.
- Exchange ratio = 137.5/150 = 0.9167.
- Meaning: maximum new Acquirer shares per Target share while maintaining positive NPV.

Q2(e)

- Exchange ratio so Target receives equity worth $5bn.
- Solution:

Q2(e)

- Exchange ratio so Target receives equity worth $5bn.

- Solution:
- Solve $x/(200+x) \times 13{,}500 = 5{,}000$.
- $x = 117.647$m shares issued.
- Exchange ratio = 117.647 / 150 = 0.7843.

Q3(a)

- Given its recent performance there is now great fear that Pleito, PLC may go bankrupt soon. The firm has assets which may be worth £100 million if things go well or £20 million if things don't go well. It is known by all market participants that the likelihood of each scenario is 50%. The firm currently has debt outstanding with face value of £50 million due next year. Please assume that there are no costs of financial distress, and the appropriate discount rate is zero.

Q3(a)

- Assets worth £100 (good) or £20 (bad), each with 50%. Debt face value £50.
- What are the current market values of the assets, equity, and debt?

- Solution:

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- Assets worth £100 (good) or £20 (bad), each with 50%. Debt face value £50.
- What are the current market values of the assets, equity, and debt?

- Solution:
- Debt value = 0.5×50 + 0.5×20 = £35m.
- Equity value = 0.5×50 + 0.5×0 = £25m.
- Enterprise value = £60m.

Q3(b)

- Project: invest £1m; payoff £5m (good), −£10m (bad).
- Solution:

Q3(b)

- Project: invest £1m; payoff £5m (good), −£10m (bad).

- Solution:
- Debt = 0.5×50 + 0.5×9 = £29.5m.
- Equity = 0.5×54 + 0.5×0 = £27m.
- Enterprise value = £56.5m.
- Equity increases → shareholders would invest.

Q3(c)

- Name of agency problem.
- Solution: Risk shifting (asset substitution).

Q3(d)

- New project: invest £10m, payoff £15m in both states (financed externally).

- Solution:

Q3(d)

- New project: invest £10m, payoff £15m in both states (financed externally).

- Solution:
- Debt = 0.5×50 + 0.5×35 = £42.5m.
- Equity = 0.5×65 + 0.5×0 = £32.5m.
- Enterprise value = £75m.

Q3(e)

- Could the firm raise £10m from equity or junior debt?

- Solution:

Q3(e)

- Could the firm raise £10m from equity or junior debt?

- Solution:
- No. Equity gains £7.5m and debt gains £7.5m, so neither would invest £10m. This illustrates the debt overhang problem.

Q4(a)

- A private company, LIME Inc. is being considered as a target by a private equity firm. The shareholders of LIME value the enterprise at $2,000 million. End of current year EBITDA is $250 million, and D&A/Capex (i.e. D&A/EBITDA) is 15%. The tax rate is 40%. The forecast holding period is 5 years. Forecast CAGR for EBITDA is 6.95% and the EV/EBITDA multiple after 5 years is expected to be10x. The financing of LIME includes senior and junior debt: Senior debt: 2.8x EBITDA ($700) at 5% interest (amortized out of annual net earnings) Junior debt: 1.6x EBITDA ($400) at 8.0% interest (with a bullet repayment).

Q4(a)

- LIME wants you to lay out the annual income statement and the annual financing of the company. In addition, show the valuation of the company now (i.e year 0) and in the year of exit.

Q4(a)

- Private equity LBO analysis.

- Solution:
- EBITDA grows from $250m to about $350m over 5 years.
- Senior debt amortizes from $700m to $85m.
- Junior debt remains $400m.
- Exit EV ≈ $3,498m.
- Equity value increases from $900m to about $3,013m.

Q4(a)

Year	0	1	2	3	4	5	
Senior Debt at end of year	700	604	495	373	237	85	
Junior debt at end of year	400	400	400	400	400	400	
EBITDA	250	267	286	306	327	350	
D&A/CAPEX		40	43	46	49	52	
EBIT		227	243	260	278	297	
Interest: senior debt		35	30	25	19	12	
Interest: junior debt		32	32	32	32	32	
EBT		160	181	203	227	253	
Taxes		64	72	81	91	101	
Net Earnings		96	109	122	136	152	
Enterprise Value	2000					3498	1.749097
Equity value	900					3013	3.348116

Q4(b)

- Calculate IRRs.
- Solution:

Q4(b)

- Calculate IRRs.

- Solution:
- Enterprise IRR ≈ 11.8%.
- Equity IRR ≈ 27.3%.

Q4(c)

- WACC = 13%. Is acquisition attractive?

- Solution:

Q4(c)

- WACC = 13%. Is acquisition attractive?

- Solution:
- Enterprise IRR (11.8%) < WACC → not attractive on enterprise basis.
- However equity IRR (27.3%) appears attractive because leverage magnifies returns. Proper comparison should match equity IRR with cost of equity.

Q4(d)

- Where have PE returns come from (Acharya & Kehoe study)?

- Solution:

Q4(d)

- Where have PE returns come from (Acharya & Kehoe study)?

- Solution:
- Returns largely driven by operational improvements (alpha), especially higher margins and productivity. Many successful deals replaced management and improved efficiency.

Key issues that we will discuss

Session 1

- Introduction to M&A (Merger vocabulary; M&A market and cast of characters; deal analysis: synergies, means of payment; valuation methods for M&A)

Session 2

- M&A performance and value creation: evidence; the value of control
- Case study: Roche-Genentech (The Swiss pharmaceutical company was a majority owner of US biotech firm Genentech from the early 90s. It is considering making a tender offer for the remaining part of Genentech that it doesn't own. What is the logic of this merger? What are the valuation and governance issues that arise?)

Session 3

- Setting deal terms, bidding tactics in M&A
- Case study: Buffett's Bid for Media General's Newspapers (Berkshire Hathaway wants to acquire Media General's newspaper division. Why is Buffett interested in this company which is in an industry in decline?)

Key issues that we will discuss

Session 4

- Regulatory aspects and advisory fees in M&A
- Financial distress, debt overhang, out-of-court restructuring

Session 5

- Case study: Preem (The refinery Corral/ Preem is in financial distress and must quickly restructure its debt out-of-court to avoid a liquidation of the business.)
- Guest Lecture

Session 6

- Distressed restructuring inside bankruptcy
- Case study: Latam Airlines (In 2020, Latam Airlines became the largest airline in the world to be driven to bankruptcy by COVID-19. With a complex debt structure and international investor composition, the company decided to file for bankruptcy protection in the U.S.)

Key issues that we will discuss

Session 7

- Different forms of ownership and governance across countries; how ownership influences control transactions, e.g., mergers; regulatory protection of minority shareholders

Session 8

- Shareholder activism by hedge funds. Who are the activists? How does it work & what are the outcomes?
- Guest lecture

Session 9

- Shareholder activism by active and passive asset managers
- Private Equity: structure of PE transactions, incentives and performance

Session 10

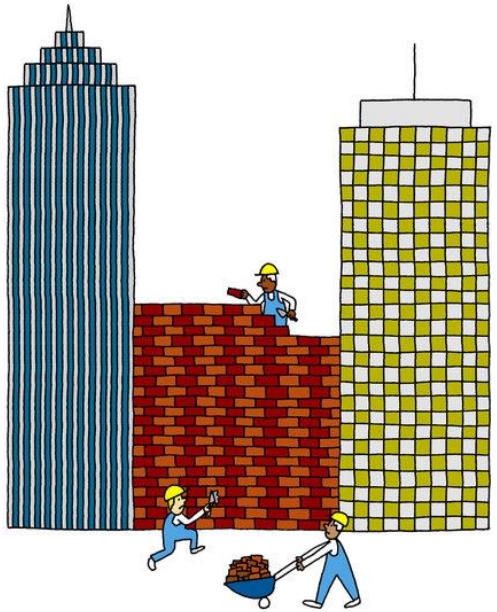
- Guest lecture
- Conglomerates, virtues and vices

Synergies: Combined firm is more valuable than the merging firms separately

- Revenue synergies
- Cost synergies
- Tax gains
- Financing gains

Agency

- Market for corporate control
- Empire building
- Overconfidence



Source: Jordan Awan, New York Times

Taxonomy of mergers

- Horizontal vs. vertical
- Related vs. unrelated (diversifying)
- Hostile vs. friendly
- Cash (fixed $ amount per share) vs. stock (fixed # of shares of acquirer)
- Industrial buyer (corporate) vs. financial buyer (PE fund)
- Single-bidder vs. multiple bidders
- Majority vs. minority

An Overview of Valuation

Mergers, MBOs & Other Corporate Reorganizations

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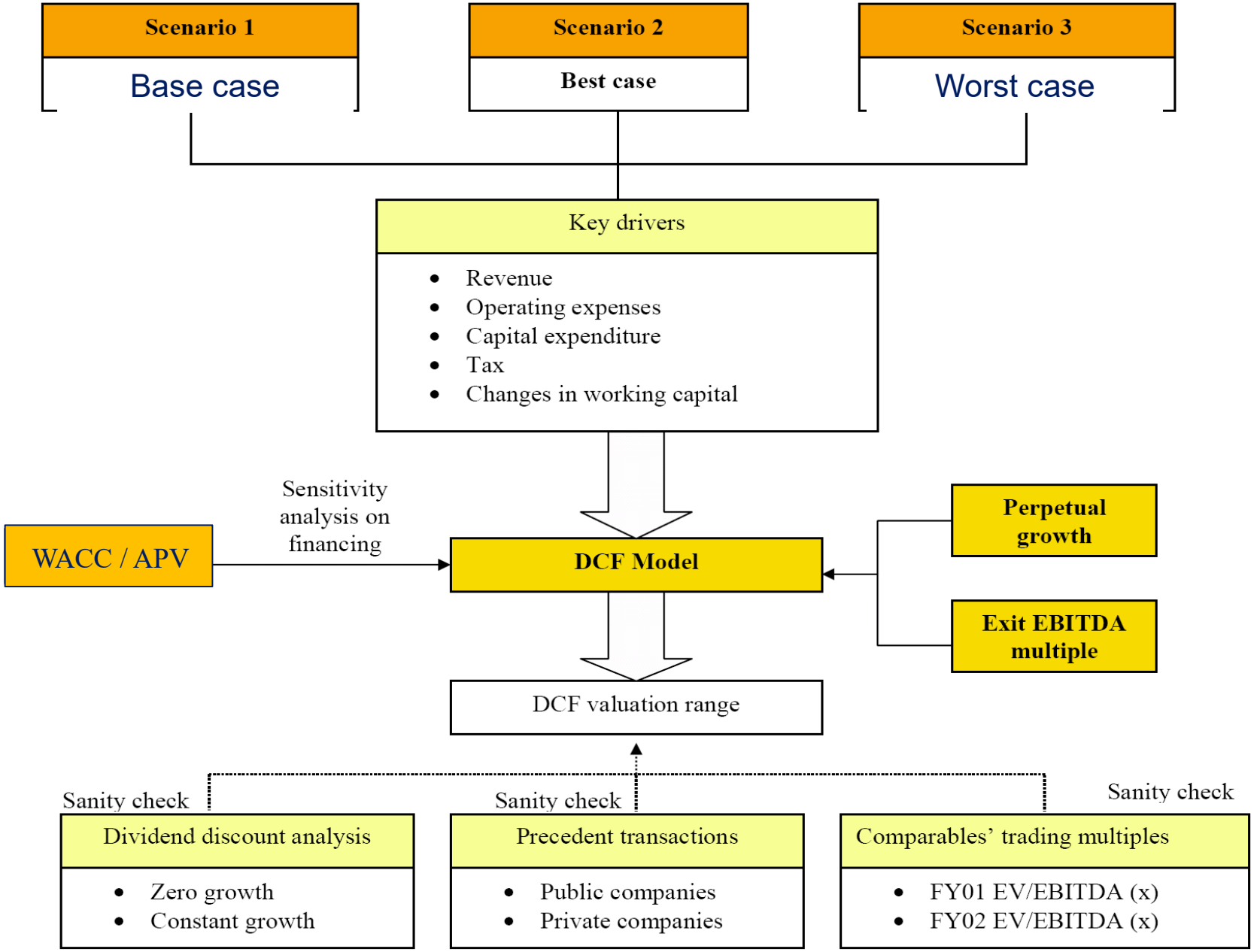

Control

- What is the “Value of Control”?
- Estimates of the value of control
- What determines the value of control?

- Calculating merger value creation from stock market data
- Merger arbitrage

- Method of payment
- How to set the Exchange Ratio in a merger
- Advanced financing provisions: Collars, CVRs, and Earnouts
- How to sell a company
- How to buy a company
- Bid tactics and bid rules
- Clauses protecting acquirers

- Bidding tactics
- Hostile takeovers

- Regulatory aspects
  - Securities law
  - Antitrust law
  - Foreign investment laws

- Financial distress
- Costs of financial distress
  - Direct
  - Indirect

- Resolving financial distress
- Debt restructuring
- Bankruptcy
  - Reorganization
  - Liquidation